

Phase locking of multiple optical fiber channels for a slow-light-enabled laser radar system

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Abstract: We have recently demonstrated a design for a multi-aperture, coherently combined, synchronized- and phased-array slow light laser radar (SLIDAR) that is capable of scanning in two dimensions with dynamic group delay compensation. Here we describe in detail the optical phase locking system used in the design, consisting of an electro-optic phase modulator (EOM), a fast 2π -phase snapback circuit to achieve phase wrapping, and associated electronics. Signals from multiple emitting apertures are phase locked simultaneously to within $1/10$ wave ($\pi/5$ radians) after propagating through 2.2 km of single-mode fiber per channel. Phase locking performance is maintained even as two independent slow light mechanisms are utilized simultaneously.

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1. Introduction

Slow light, or light that travels at extremely slow group velocities, has emerged not only as an intriguing area of fundamental science but also with promise as an important enabling technology for diverse applications, especially in optical fibers [1–5]. One of the hallmarks of slow light technology is the ability to tune the group velocity (and likewise the group delay) controllably and repeatedly, and this tunability is at the heart of many proposed and demonstrated applications [6–10].

We have recently demonstrated the applicability of slow light to the design of a multi-aperture coherently combined pulsed laser radar system [11, 12]. The use of tunable slow light delay lines overcomes the fundamental limitation of group delay mismatch during wide-angle beam steering, opening up the possibility of high-performance laser radar with both short pulses and wide steering angles as well as good transverse and longitudinal resolution. Such systems require both tunable slow light delay elements and techniques for phase locking the emitter array. For a pulsed, phased-array laser radar with short enough pulses, a long enough emitter baseline, and a wide enough field of view, a tunable true time delay is required to synchronize the arrival of pulses from each emitter in the far field; two slow light methods provide that tunable true time delay in our system. To achieve constructive interference, the emitters' signals must be coherent and properly phased, necessitating the use of a phase locking mechanism. Indeed, the use of a phased array of emitters is conceptually similar to coherent beam combining in its requirement of phase control [13–17].

We present here the phase locking techniques used in our slow light laser radar (SLIDAR) system. The phase locking system satisfies the need for a simple, inexpensive, yet reasonably effective phase control system to allow the proper phasing of each emitter. In particular, the use of an electro-optic phase modulator (EOM) and a single-quadrature phase detector in a heterodyne configuration allows phase locking of the optical signals to within a tenth of a wave RMS ($\pi/5$ radians), and a fast 2π -phase snapback circuit overcomes the problem of finite phase actuation range. While several quite versatile phase locking techniques have been developed, such as optical phase-locked loops (OPLL) [16–18] and LOCSET [19, 20], our slow light system precludes the use of tuning the source laser wavelength, as in an OPLL, and requires that all signal channels operate at the same frequency, unlike LOCSET. Further, the phase locking system presented here does not require the complex signal processing electronics used by such feedback control methods as stochastic parallel gradient descent (SPGD) optimization [21–23].

2. Approach

The optical system employs two independent slow light mechanisms, namely dispersive delay [3] and stimulated Brillouin scattering (SBS) slow light [5, 24]. By controlling the wavelength of the optical field and the pump power of the SBS module, one can achieve independent group delay compensation in two orthogonal transverse dimensions. Details of the group delay modules have been reported previously [11, 12].

For phase control, a small portion of each channel’s output signal is split off and mixed with a 55-MHz-shifted optical reference field. The detected heterodyne beat signal is sent to a phase locking circuit, which feeds back to an electro-optic phase modulator in each channel to control the relative optical phase of the emitted fields.

2.1. Optical System

The layout of the optical system is shown in Fig. 1. A fiber-coupled tunable laser (TL) generates about 1 mW of continuous-wave (cw) light near 1550 nm and is protected by an optical isolator. The laser feeds both a phase reference arm and a signal arm. The signal arm contains a pulse carver followed by multiple signal channels. The pulse carver consists of an intensity modulator (IM) driven by an arbitrary function generator (AFG) as well as fiber polarization controllers and an erbium-doped fiber amplifier (EDFA). Each signal channel contains a 2.2-km length of single-mode fiber (SMF), either dispersion-compensating fiber (DCF) or dispersion-shifted fiber (DSF). An electro-optic phase modulator (EOM) adjusts the phase of the signal channel to maintain phase lock, and a final EDFA operating in constant-output mode delivers the output signal at a constant power level to the emitting aperture and the heterodyne detection setup. The phase reference is frequency-shifted by an acousto-optic modulator (AOM) driven at 55 MHz by a fixed radio frequency (RF) driver. The frequency-shifted phase reference is heterodyned with a fraction of the signal channel output to produce a beat signal for phase locking. Only one channel is shown in Fig. 1, but in our system, several channels with different slow light mechanisms are phase-locked to the same phase reference. Our system uses three signal channels, with the three emitters in an L-shaped arrangement (shown in the inset of Fig. 1) that allows two-dimensional steering. (Note that the short length of fiber in each signal channel between the last splitter and the emitter cannot be monitored for phase noise; this uncompensated fiber length is kept as short as possible to minimize the additional phase noise it contributes to the system.)

The pulse carver in the signal arm creates 6.5 ns pulses (FWHM) on a cw background of about 30% of the peak power. The pulse duration is shorter than the response time of the phase control electronics, so the phase control circuit does not “see” the pulse. Only the cw background is used by the phase control electronics. Each signal channel is kept in phase with the

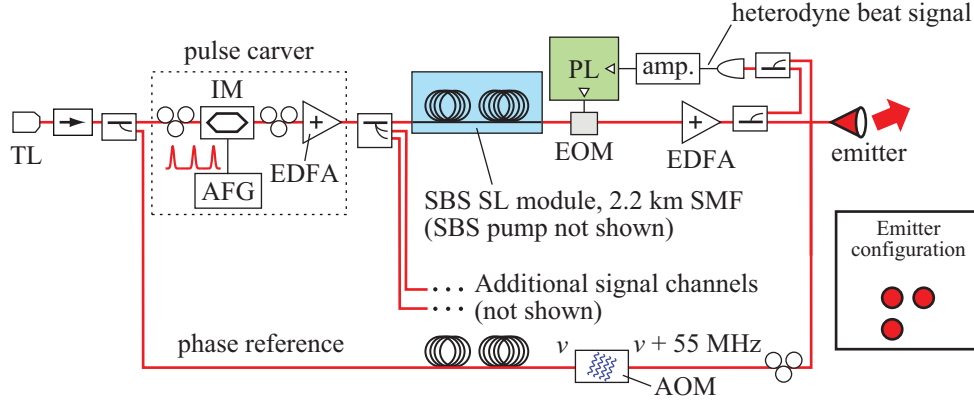


Fig. 1. Schematic diagram of the optical system (see text for details). TL: tunable laser; IM: intensity modulator; AFG: arbitrary function generator; SBS SL: stimulated Brillouin scattering slow light; SMF: single-mode fiber; EOM: electro-optic phase modulator; PLL: phase-locked electronics; EDFA: erbium-doped fiber amplifier; AOM: acousto-optic modulator; amp.: RF amplifier. For simplicity, only one signal channel is shown here. Inset: Emitter configuration, as seen facing into the emitters.

phase reference, allowing pulses from multiple channels to be combined coherently.

Tuning the pulse delay using dispersive slow light does not affect the phase control system, as a change in wavelength does not in itself change the heterodyne beat signal. However, tuning the SBS delay changes the gain experienced by the signal. A fluctuating heterodyne signal amplitude would make it impossible to maintain phase lock, so the variable gain of the SBS process is compensated by the final EDFA. This EDFA operates in constant-output mode, not in saturation, so it keeps the cw background (and therefore the heterodyne signal) at a constant amplitude. Further, the pulse is shorter than the response time of the EDFA's control electronics in this mode, so the EDFA does not adjust its gain level in response to the pulse (which would distort the pulse).

2.2. Electronics

A block diagram of the phase control electronics is shown in Fig. 2. The interference of the signal channel's cw background with the frequency-shifted phase reference signal creates a heterodyne beat signal centered at 55 MHz. The detected heterodyne beat signal is amplified and then passes through phase detection electronics that produce a baseband phase error signal. Specifically, the beat signal is mixed with a phase-shifted version of the 55 MHz RF local oscillator (LO), then low-pass filtered. (Note that the same LO is used to drive the frequency-shifting AOM in the phase reference arm. Having a phase shifter in each channel's electronics allows control of the relative phase of each signal channel.) A proportional-integral (P-I) controller, consisting of a loop filter with gain and a fast 2π -phase snapback circuit, filters the error signal and drives the EOM. The design of the P-I controller is shown in Fig. 3. PSpice circuit simulations and laboratory measurements of the loop filter transfer function show the loop filter bandwidth to be about 2.3 MHz.

The phase error accumulates rapidly in the P-I controller, and the control signal to the EOM must be kept within the supply voltages of the loop filter. To accomplish this goal, a fast 2π -phase snapback circuit was designed and implemented. Its function is to rapidly add or subtract 2π radians (a full cycle of phase) from the phase error being tracked by the P-I controller whenever the controller's output voltage approaches a voltage supply level. In this way, the

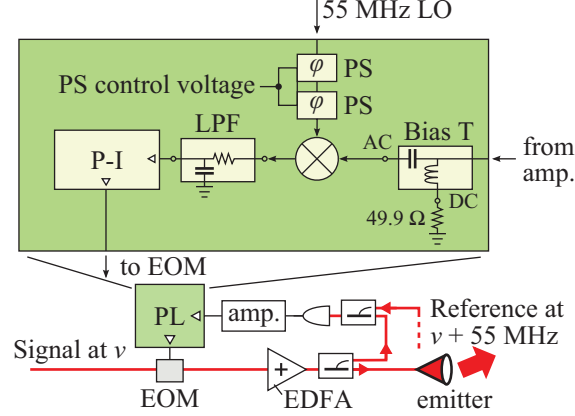


Fig. 2. Block diagram of the phase control electronics. LO: local oscillator; amp.: RF amplifier; LPF: low-pass filter; PS: RF phase shifter; P-I: proportional-integral; EOM: electro-optic phase modulator; EDFA: erbium-doped fiber amplifier. The detector is a Thorlabs model DET01CFC fiber-coupled, biased InGaAs detector. The RF amplifier is a Stanford Research model SR440 DC-300 MHz amplifier. The bias T, PS, mixer, and LPF are respectively Mini-Circuits models ZFBT-4R2G+, JSPHS-51+, ZEM-2B+, and SLP-1.9+. Two phase shifters were used to ensure more than 2π of phase tuning range.

P-I controller is able to track an unbounded amount of phase error. These brief phase snapback events do not impact performance as long as the snapback duty cycle (or frequency of snapback events) is kept low [25].

The operation of the snapback circuit is as follows: The half-wave voltage (V_π) of our EOM is nominally 4 V, and the loop filter uses ± 15 V supply voltages. When the loop filter's output V_{EOM} exceeds $3V_\pi$ (nominally 12 V), an analog switch engages and strongly drives V_{EOM} toward negative voltages. Once V_{EOM} is below V_π (nominally 4 V), the analog switch disengages, and the loop filter resumes tracking, having shifted its operating point by 2π radians of phase error.

Several issues of detail are illustrated in Fig. 3. In our design, the loop filter is an inverting circuit (negative voltage gain), so to drive V_{EOM} negative, the analog switch connects the positive voltage supply to the loop filter's input summing junction. The analog switch disengages the snapback by connecting signal ground to the input summing junction; adding 0 V to the summing junction in this way is conceptually the same as disconnecting the snapback circuit from the loop filter, but the signal inputs are not left floating. The logic of engaging and disengaging the analog switch is handled by two analog comparators and a set-reset (S-R) latch. Careful attention must be paid to the logic sense of each input or output (normal or inverted). A complementary circuit (not shown in Fig. 3) handles negative-voltage snapbacks, triggering when V_{EOM} goes below $-3V_\pi$ and disengaging when V_{EOM} returns above $-V_\pi$. Allowing the phase error to wander over a range of $\pm 3\pi$ before activating the snapback reduces the frequency of snapback events, giving a lower residual phase error than a naive approach that simply tracks the phase error modulo 2π .

The entire snapback process takes about 1 μs . Since this is longer than the response time of the loop filter, the circuit parameters must be adjusted to produce an optimal snapback effect. Specifically, the DC voltage levels used by the analog comparators and the value of the resistor that determines the snapback circuit drive strength were tuned empirically. These values may also be chosen to cause a phase snapback of integer multiples of 2π radians. In our system, the snapback circuit was tuned to provide a 4π snapback. We found that the phase error tended to drift continuously in one direction (either positive or negative), presumably due to environ-

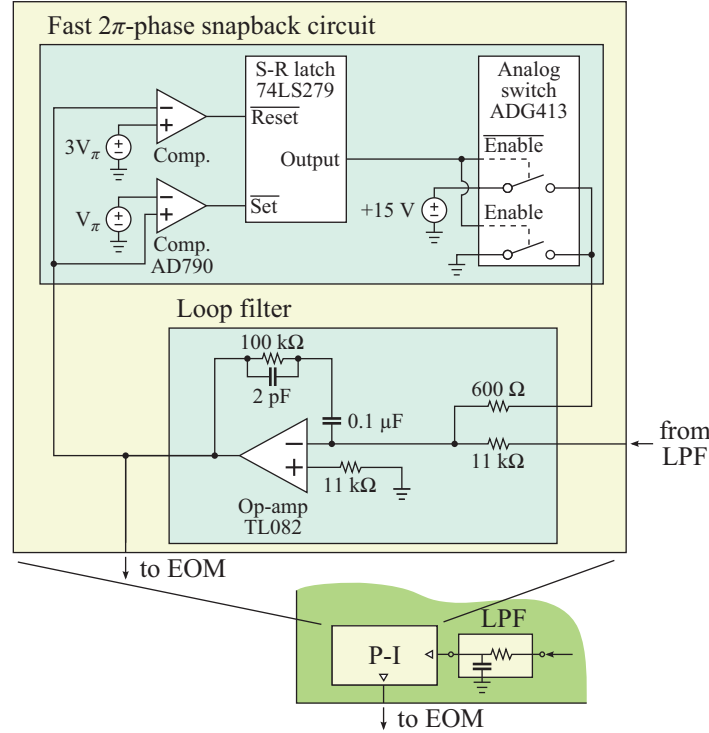


Fig. 3. Block diagram of the proportional-integral (P-I) controller, including the loop filter and the fast 2π -phase snapback circuit. EOM: electro-optic phase modulator; LPF: low-pass filter; Comp.: analog comparator (output is logical 1 when voltage at positive input exceeds voltage at negative input). Bars over input names indicate inverted-sense (active-low) logic inputs. Note that only the positive-voltage arm of the snapback circuit is shown here; a complementary circuit provides negative-voltage snapback functionality.

mental temperature changes, and the 4π snapback allowed a longer time between snapback events.

3. Results

An important figure of merit for the phase locking circuit is the RMS residual phase error, or the degree to which the heterodyne beat signal and the local oscillator are kept in a fixed phase relationship. (In our system, the phase locking circuit only needs to maintain a fixed phase relationship; any constant phase offset is compensated by the RF phase shifters in each signal channel.) An ideal phase locking circuit with infinite response bandwidth would keep the LO and heterodyne signal in perfect phase lock at all times, and there would be zero jitter (time-dependent fluctuation) in the phase between the LO and heterodyne signal. In practice, the finite response bandwidth of the system means that there will always be some fluctuating residual phase error.

The RMS residual phase error can also be used to predict the efficiency of a coherent beam combining system. The Strehl ratio S of the system is defined as the ratio of the expected value of the central far-field lobe intensity (in the presence of some residual phase error) to its theoretical peak intensity (with no residual phase error). A Strehl ratio of 1 is achieved by a system with zero RMS residual phase error, while increasing residual error will lead to a

reduced Strehl ratio and reduced far-field intensity. As such, the Strehl ratio serves as a good measure of the efficiency of coherent beam combination.

Assuming the residual phase error in each of the three emitters in our system is uncorrelated and has a Gaussian distribution with zero mean and RMS value (standard deviation) of σ radians, it can be shown that the Strehl ratio is

$$S = \frac{1}{3} + \frac{2}{3} \exp(-\sigma^2). \quad (1)$$

Note that, as expected, zero residual phase error gives $S = 1$, and for large residual phase error, $S \rightarrow 1/3$, the limit of incoherent combination. The derivation of Eq. (1) is similar to the derivation of Eq. (13) given in Ref. [13]. The residual phase error is taken to have zero mean, without loss of generality; any mean residual phase error is a constant phase offset that is compensated by the RF phase shifters.

Figure 4 shows the phase noise measured in 2.2 km of SMF over 8 ms and the residual phase error after phase locking. The phase noise in our system is believed to be caused primarily by temperature fluctuations and mechanical vibrations [25]. A snapback event is evident as a spike in the residual error halfway through the data, shown as a phase ramp as the snapback circuit shifts the phase by 4π .

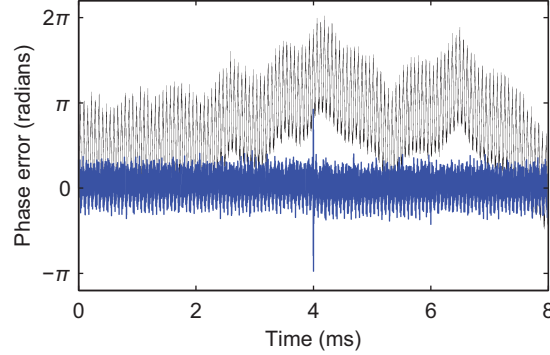


Fig. 4. Phase noise (thin black line) and residual error after correction by the phase control system (thick blue line) over a duration of 8 ms. The narrow spike in the residual phase error at 4 ms is due to a phase snapback event. The residual error is shown modulo 2π radians; the phase error is shown without the modulus.

Over the short time window shown in Fig. 4, the residual RMS phase error is just under $\pi/8$ radians, corresponding to a theoretical Strehl ratio of 0.91. The single snapback event accounts for only about 0.21% of the residual phase error. However, as one might expect, the value of both the RMS phase noise and the RMS residual phase error increase as the mean is taken over an increasing time window [25]. A more realistic estimate of the RMS residual error is given in Fig. 5, where averages were taken over about 20 s. Over this longer time window, the RMS residual error is approximately $\pi/5$ radians (1/10 wave), giving a Strehl ratio of 0.78, comparable to results obtained in several other optical phase locking systems [25, 26].

Figure 5 demonstrates the consistent phase locking performance of the system in the presence of variable SBS gain. The data in Fig. 5 show the RMS residual phase error in a counter-pumped SBS slow light configuration (using dispersion-shifted fiber) for different values of the SBS gain. The final EDFA in each signal channel ensures a constant heterodyne signal level for all levels of SBS gain. Each of the insets of Fig. 5 is similar to an eye diagram, in that it shows many overlaid traces of the heterodyne beat signal. The traces were recorded with

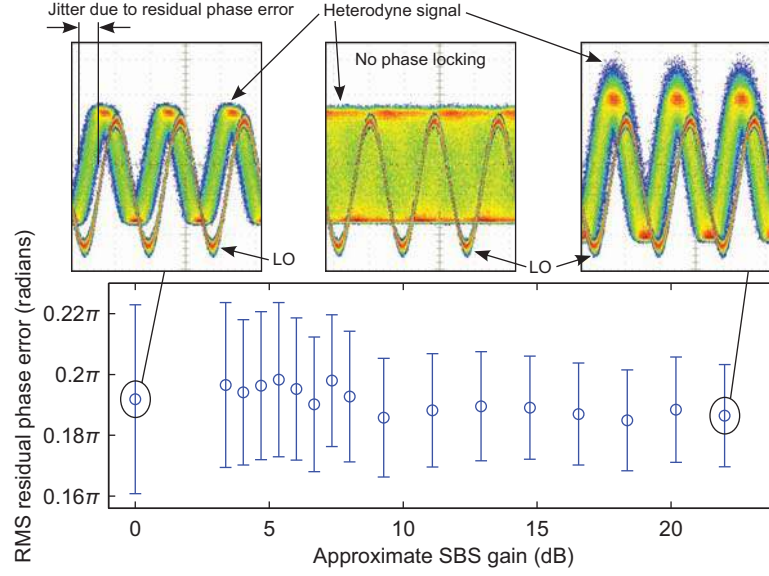


Fig. 5. RMS residual phase error vs. approximate SBS gain. The phase locking system performed equally well under all configurations. Data in the center were taken with the SBS pump broadened, and data on the right were taken with the pump unbroadened. Insets: Eye diagrams of the local oscillator (thin, well-defined sinusoidal trace) and the heterodyne beat signal (blurry trace) for various configurations; left inset: with no SBS pump power; right inset: with maximum SBS pump power; middle inset: with phase locking turned off, and therefore no fixed phase relation between LO and heterodyne signal.

the oscilloscope triggering on the LO (thin sinusoidal signal). The width or “blurriness” of the heterodyne signal is an indication of the residual phase error. The SBS gain varies from zero (no gain) to 22 dB, and the circuit maintains phase lock throughout this range. For unpumped dispersion-compensating fiber, the observed performance is similar to that shown in Fig. 5 for the unpumped (0 dB) DSF, and phase locking performance was observed not to change as the optical signal wavelength changes. The phase locking quality is the same regardless of the type or magnitude of the slow light effect. In other words, the phase lock is maintained in the presence of both SBS slow light and dispersive slow light over the full range of operation.

Figure 6 shows the far-field intensity for three signal channels with and without phase locking. The intensity oscillates over full range before the phase locking circuits are engaged, but the phase locking circuit increases the coherence once engaged. (For this particular data set, the far-field detector was located at a null in the far-field pattern, rather than at the peak of a lobe; of course, the locations of the peaks and nulls can easily be shifted by tuning the relative phase offsets of the three channels.) Figure 7 shows a real-time video of the far-field pattern with and without phase locking.

4. Conclusion

We have demonstrated a simple, effective phase locking system using an electro-optic phase modulator and a feedback circuit with a fast 2π -phase snapback, and we have shown that it is effective in maintaining phase lock among three slow light channels consisting of 2.2 km of single-mode fiber. The RMS residual phase error is kept below $\pi/5$ radians, and the relative phases of the signal channels can be adjusted arbitrarily. The proportional-integral controller

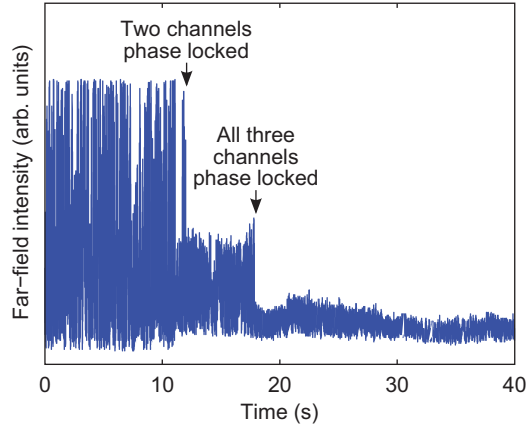


Fig. 6. Far-field intensity for three signal channels, with and without phase lock. Full-range oscillations on the left indicate a lack of phase lock, while reduced fluctuations indicate a stable phase lock.

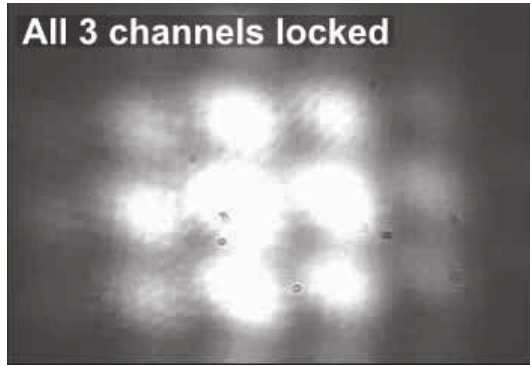


Fig. 7. (Multimedia online) A representative frame from a real-time video showing the far-field pattern of the three-channel system with and without phase locking.

provides adequate voltage to drive the EOM over a range of more than 6π radians without additional amplification. The phase locking approach is scalable to many more channels, as each channel is independently locked to an optical reference. The approach presented here is effective for slow light methods in optical fibers several kilometers long and should also be applicable in coherent beam combining, where fiber lengths are typically limited to tens of meters.

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